Climate and Environment Group 3: Carbon Pricing

Qingyue Chen, Puran Dou, Yi Dou, Gavin Gondalwala, Catherine Morris

#1 Topic and Motivation

Policy Area and Problem

This research investigates the structural drivers of carbon leakage, specifically bifurcating the impact of intermediate supply chains versus final-goods trade. Carbon leakage, or the cross-border migration of CO2 emissions, occurs when stringent domestic carbon pricing incentivizes firms to relocate production to jurisdictions with more permissive regulatory frameworks.

While this shift may result in a nominal decrease in local emissions, it frequently results in negative externalities, leaving global aggregate emissions unchanged or, in cases of lower production efficiency abroad, net-increased. Our analysis aims to determine if price-induced leakage is driven primarily by supply chain inputs (intermediate goods and raw materials) or the trade of goods.

Policy Relevance and Magnitude

Distinguishing which phase of the production cycle is driving carbon price leakage is critical for effective climate interventions. The global carbon pricing landscape is expanding: as of 2025, carbon pricing mechanisms cover approximately 28% of global greenhouse gas emissions, a significant increase from less than 5% in 2005. Furthermore, the 55 jurisdictions with active carbon taxes or Emissions Trading Systems (ETS) now represent nearly two-thirds of global GDP.

Despite this expansion, contemporary policy instruments—such as the Carbon Border Adjustment Mechanism (CBAM)—largely target carbon-intensive final goods like steel and cement. If the data suggests that significant leakage is occurring deeper within the supply chain, this suggests a misalignment within current policy frameworks, undermining the effectiveness of carbon pricing strategies.

Who Is Affected and Why New Evidence Matters

CO2 is a primary driver of global warming, meaning that it is not too hyperbolic to say that everyone on Earth is affected by carbon leakage. If carbon emissions policies fail to effectively rein in CO2 emissions—or inadvertently cause emissions to increase—they may simply be a form of greenwashing, nice on paper, but failing to meaningfully slow the rise in global temperatures. Understanding which point in the production cycle is having the greatest impact on emissions can help nations craft more effective policies.

#2 Research Question

Research Question

Do carbon pricing policies cause carbon leakage primarily through upstream input supply chains rather than through final-goods trade flows?

Key Variables

Treatment / Exposure (X): The introduction or change in carbon pricing policy (e.g., carbon tax or emissions trading system) in a given jurisdiction.

Outcome (Y): Changes in carbon leakage, measured as:

- Increased embodied emissions in imports, and/or
- Changes in trade volumes in intermediate goods versus final goods across borders.

Population: Carbon-intensive industries engaged in international trade, particularly sectors with complex global supply chains (e.g., steel, cement, chemicals, and manufacturing sectors) across countries with and without carbon pricing mechanisms.

#3 Theory of Change

A. Market Failure

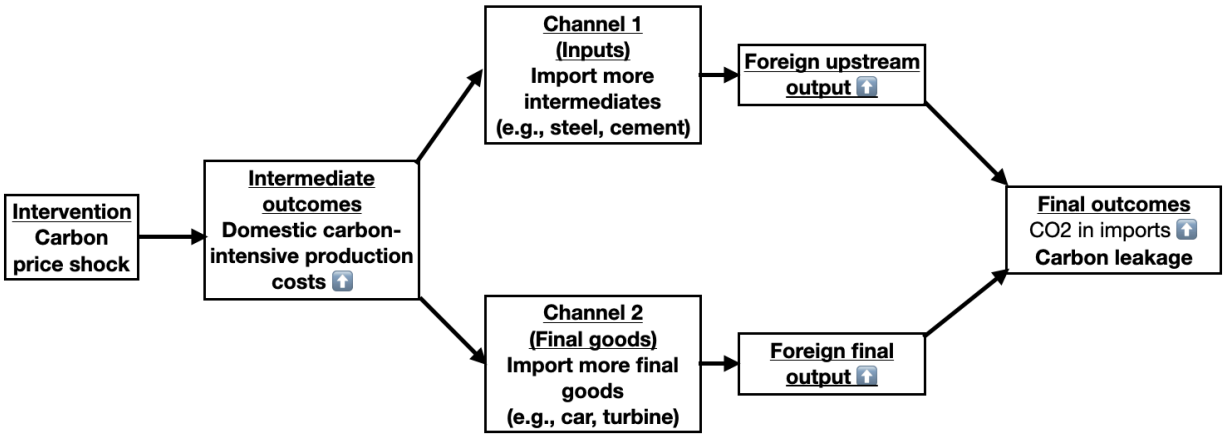
Climate change represents a classic case of a **global negative externality**. Greenhouse gas emissions impose substantial social costs that are not fully borne by the firms or countries responsible for them. As a result, in the absence of policy intervention, carbon-intensive production is underpriced relative to its true social cost, leading to excessive global emissions. Carbon pricing mechanisms are designed to internalize this externality by raising the cost of emitting carbon and aligning private incentives with social welfare. However, because climate policy is not globally harmonized, carbon pricing is often implemented unilaterally or by a subset of countries. This creates **asymmetric regulatory environments** in which firms in regulated jurisdictions face higher production costs than competitors in unregulated regions. Such cost differentials can distort production and trade patterns, as firms seek to maintain profitability by shifting production, sourcing intermediate inputs from abroad, or relocating investment to countries with weaker climate policies. In this context, emissions may decline domestically but increase elsewhere—a phenomenon known as carbon leakage.

Building on this framework, carbon leakage is unlikely to be uniform across sectors and may depend systematically on the positions in a supply chain. In particular, upstream industries—such as energy, metals, cement, and primary chemicals—are typically more carbon-intensive and face direct carbon cost exposure under pricing schemes. Because emissions constitute a significant share of their marginal production costs, in these upstream sectors, even moderate cost differentials can shift **comparative advantage** across jurisdictions. Firms in these sectors face strong incentives to relocate production, substitute toward foreign suppliers, or redirect exports and imports in response to asymmetric carbon pricing.

By contrast, downstream industries are generally less directly carbon-intensive and are affected primarily through input cost transmission rather than direct emissions pricing. Their total production costs reflect a broader mix of labor, capital, technology, branding, and product differentiation, which may dampen the relative importance of carbon costs in shaping location decisions. Moreover, downstream goods are often more differentiated and embedded in complex supply chains, reducing the **elasticity of substitution** across countries. As a result, while carbon pricing can still affect downstream sectors indirectly, the market mechanism for leakage is theoretically stronger in upstream stages of production where carbon costs are more salient and internationally comparable.

B. The Casual Chain

- **Draw a simple DAG or logic model**



- **Intervention → Intermediate outcomes → Final outcomes**
  - **Intervention:** carbon price shock (introduction or increase of a carbon tax, or tightening in ETS carbon allowances)
  - **Intermediate outcomes:**
    - Domestic carbon-intensive production costs rise.
    - Import sourcing shifts toward jurisdictions with weaker carbon constraints / lower carbon prices.
    - Channel 1: Input/supply-chain adjustment. Firms import more carbon-intensive intermediate inputs and/or offshore carbon-intensive stages.
    - Channel 2: Final-goods adjustment. Domestic demand shifts toward imported final goods (import penetration increases).
  - **Final outcomes:** Carbon leakage occurs, foreign emissions rise and offset part of domestic reductions. Global net abatement is lower than domestic abatement.

- **Identify key assumptions at each step & What mechanisms link your intervention to outcomes**
  - **Overall assumption:**
    - Carbon price shock raises marginal cost of carbon-intensive domestic production, pushing firms to adjust accordingly to reduce their cost exposure.
    - Firms adjust through two trade-related channels: upstream input-sourcing shifts and final-goods imports increases.
  - **Intervention → Domestic cost increase:** the shock is large and persistent enough for enterprises…
  - **Domestic cost increase → Firms choose trade/sourcing adjustments: …** to re-optimize sourcing and location choices.
  - **Domestic cost increase → Channel 1 (upper value chain):** supply chains can be reconfigured as enterprises intended
  - **Domestic cost increase → Channel 2 (final goods):** substitutes for final goods are available, buyers are price-responsive and accept import goods
  - **Intermediate outcomes → Leakage (final outcomes):** import are concentrated in lower carbon regulation and/or higher emissions intensity origins

Initial Literature Scan

The proliferation of unilateral climate policies has necessitated a deeper understanding of carbon

leakage, the displacement of emissions to jurisdictions with weaker regulations. Recent scholarship suggests that while trade in final goods accounts for a significant volume of embodied carbon, the mechanisms driving leakage are disproportionately concentrated in upstream, energy-intensive input sectors.

Current policy designs reflect the tension between regulating upstream inputs and managing downstream trade slows. To mitigate leakage, jurisdictions such as the EU have historically utilized "free allocation of allowances." Calel and Dechezleprêtre (2016) find that under the EU ETS regulated firms (primarily power stations and industrial plants) did respond to proving through low-carbon innovation and operational changes like fuel switching, However, Grubb et al, (2022) argue that shielding upstream sectors dampens the price signal necessary for downstream efficiency, therefore raising overall decarbonization costs.

This has led to proposals for Border Carbon Adjustments (BCAs). Böhringer et al. (2022) argue that BCAs effectively curb leakage by leveling the field for emissions-intensive trade-exposed (EITE) industries. A critical risk regarding the scope of these policies, however, is that if BCAs only target upstream raw materials without covering downstream intermediate goods, leakage may simply be "diverted" through imports of other goods.

While much of this literature has focused on the dichotomy of upstream and downstream goods, our research seeks to identify the intermediate levels where leakage is most likely to occur. Building on Böhringer et al.'s ideas, effective policy can only be designed with clear targets in mind.

Böhringer, Christoph, Carolyn Fischer, Knut Einar Rosendahl, and Thomas Fox Rutherford. 2022. "Potential Impacts and Challenges of Border Carbon Adjustments." *Nature Climate Change* 12: 22–29. https://doi.org/10.1038/s41558-021-01250-z.

Calel, Raphael, and Antoine Dechezleprêtre. 2016. "Environmental Policy and Directed Technological Change: Evidence from the European Carbon Market." *Review of Economics and Statistics* 98 (1): 173–191. https://doi.org/10.1162/REST_a_00470.

Grubb, Michael, Nino David Jordan, Edgar Hertwich, Karsten Neuhoff, Kasturi Das, Kaushik Ranjan Bandyopadhyay, Harro van Asselt, Misato Sato, Ranran Wang, William A. Pizer, and Hyungna Oh. 2022. "Carbon Leakage, Consumption, and Trade." *Annual Review of Environment and Resources* 47: 753–795. https://doi.org/10.1146/annurev-environ-120820-053625,